

Dynamic Guideline: WHO-Based Obesity - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Scope note: This report synthesizes **exactly 5 WHO guidelines** directly relevant to obesity diagnosis, prevention, treatment, or management. The WHO document titled “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” is **not an obesity guideline**; it is a youth-engagement case study linked to the WHO Abortion care guideline, although its principles are relevant to future adolescent-obesity guideline development [WHO webinar](#).

No.	WHO Guideline	Publication Date	Target Population	Scope and Objectives	Status as of 29 April 2026
1	WHO guideline on the use of glucagon-like peptide-1 therapies for the treatment of obesity in adults	1 December 2025	Adults living with obesity	Safe, equitable, appropriate use of GLP-1 receptor agonists and GIP/GLP-1 dual agonists as part of comprehensive chronic obesity care, not as standalone treatment WHO news release , PAHO/WHO	Latest final WHO obesity-treatment guideline
2	Total fat intake for the prevention of unhealthy weight gain in adults and children: WHO guideline	17 July 2023	Adults and children	Dietary fat quantity; prevention of unhealthy weight gain; total fat ≤30% of total energy intake in adults WHO news , NCBI Bookshelf	Final
3	Saturated fatty acid and trans-fatty acid intake for adults and children: WHO guideline	17 July 2023	Adults and children	Dietary fat quality; reduce saturated fatty acids and trans-fatty acids to lower diet-related NCD risk, especially cardiovascular disease WHO guideline	Final

No.	WHO Guideline	Publication Date	Target Population	Scope and Objectives	Status as of 29 April 2026
4	Use of non-sugar sweeteners: WHO guideline	15 May 2023	General population, excluding people with pre-existing diabetes for diabetes management purposes	Guidance on whether non-sugar sweeteners should be used to reduce free-sugar intake, prevent unhealthy weight gain, and reduce diet-related NCD risk WHO non-sugar sweeteners guideline	Final
5	Sugars intake for adults and children: WHO guideline	4 March 2015	Adults and children	Reduce free-sugars intake to lower risk of unhealthy weight gain, obesity-related NCDs, and dental caries WHO sugars intake guideline	Final

WHO diagnostic definitions relevant to obesity

Population	WHO definition
Adults	Overweight: BMI ≥ 25 kg/m² ; obesity: BMI ≥ 30 kg/m² WHO sugars intake guideline
Children under 5 years	Overweight: weight-for-height > +2 SD of WHO Child Growth Standards median WHO sugars intake guideline
Children and adolescents aged 5–19 years	Overweight: BMI-for-age > +1 SD ; obesity: BMI-for-age > +2 SD of WHO reference WHO BMI-for-age reference

1.2 Key Recommendations

GRADE framework used in this report

GRADE domain	Interpretation
High	Further research is very unlikely to change confidence in the estimate of effect.
Moderate	Further research may change confidence in the estimate.
Low	Further research is likely to change confidence in the estimate.
Very Low	Any estimate of effect is very uncertain.

GRADE domain	Interpretation
Strong recommendation	Most individuals should receive the recommended intervention or policy.
Conditional recommendation	Different choices may be appropriate depending on patient values, feasibility, equity, cost, resources, and context.

WHO Guideline	Recommendation	GRADE Evidence Quality	Strength
GLP-1 therapies for adults with obesity	Adults living with obesity may use GLP-1 therapies as long-term treatment within comprehensive chronic obesity care JAMA/WHO guideline PDF copy , PubMed	Moderate	Conditional
GLP-1 therapies for adults with obesity	Adults prescribed GLP-1 therapies may receive intensive behavioral therapy as part of a multimodal clinical algorithm JAMA/WHO guideline PDF copy	Low	Conditional
GLP-1 therapies for adults with obesity	Provide context-appropriate counseling on healthy diet, physical activity, and behavioral change as the first step toward structured care JAMA/WHO guideline PDF copy	Not formally GRADE-rated; good-practice statement	Good practice
Total fat intake	Adults should limit total fat intake to ≤30% of total energy intake to help prevent unhealthy weight gain NCBI Bookshelf	Low	Conditional
Total fat intake	Fat intake should be considered alongside total energy balance, free sugars, carbohydrate quality, non-sugar sweeteners, energy requirements, and physical activity NCBI Bookshelf	Low	Conditional
Saturated and trans-fatty acids	People aged ≥2 years should consume fat primarily as unsaturated fatty acids , with saturated fatty acids ≤10% of total energy intake and trans-fatty acids ≤1% WHO news , WHO guideline	Moderate	Strong
Saturated and trans-fatty acids	Replace saturated and trans fats with polyunsaturated fats, plant-source monounsaturated fats, or carbohydrates from high-fibre foods such as whole grains, vegetables, fruits, and pulses WHO news	Moderate	Strong
Non-sugar sweeteners	Do not use non-sugar sweeteners as a strategy for long-term weight control or reduction of diet-related NCD risk in the general population; short-term RCT weight effects are	Low	Conditional

WHO Guideline	Recommendation	GRADE Evidence Quality	Strength
	outweighed by uncertainty and possible long-term adverse associations WHO non-sugar sweeteners guideline		
Sugars intake	Reduce free sugars to <10% of total energy intake in adults and children WHO sugars intake guideline , WHO healthy diet fact sheet	Moderate	Strong
Sugars intake	Further reduce free sugars to <5% of total energy intake where feasible for additional health benefits WHO healthy diet fact sheet , WHO obesity discussion paper	Low	Conditional

Integrated WHO-based clinical interpretation

1. **Diagnose obesity using BMI and age-specific WHO growth references**, but assess health risk more broadly, including comorbidities, central adiposity, stigma, functional impairment, and social determinants.
2. **Treat obesity as a chronic, relapsing disease**, not as an isolated lifestyle failure. WHO's 2025 GLP-1 guideline explicitly frames obesity as requiring lifelong care [PAHO/WHO](#).
3. **Dietary management should address both quantity and quality**:
 - Total fat $\leq 30\%$ of energy.
 - Saturated fat $\leq 10\%$ and trans fat $\leq 1\%$.
 - Free sugars $< 10\%$, preferably $< 5\%$ where feasible.
 - Replace unhealthy fats and refined carbohydrates with unsaturated fats and fibre-rich foods.
4. **Non-sugar sweeteners should not be relied on for long-term obesity prevention or weight control** because WHO found mixed evidence: short-term RCTs showed lower weight/BMI, but long-term observational studies associated higher intake with increased BMI and incident obesity [WHO non-sugar sweeteners guideline](#).
5. **Pharmacotherapy should be multimodal**: GLP-1 or GIP/GLP-1 therapy may be used for adults with obesity, but only within chronic-care systems that include behavioral support, nutrition care, physical activity, monitoring, adverse-event management, affordability, and equity safeguards [WHO news release](#).

1.3 Guideline Development Methodology

Methodological component	WHO approach across the 5 guidelines
Evidence review process	WHO used systematic evidence reviews, expert guideline-development groups, and evidence-to-decision processes. The GLP-1 guideline synthesized efficacy, effectiveness, and safety evidence for GLP-1 receptor agonists and GIP/GLP-1 dual agonists PAHO/WHO .
GRADE	WHO used the GRADE approach for recommendations in the GLP-1 guideline and dietary guidelines WHO guideline PDF via MAGICapp , NCBI Bookshelf .

Methodological component	WHO approach across the 5 guidelines
GRADE-CERQual	The GLP-1 guideline also used GRADE-CERQual for qualitative evidence on values, preferences, acceptability, and equity impacts PAHO/WHO .
Consensus method	Recommendations were developed through multidisciplinary guideline-development group deliberation using evidence-to-decision frameworks, including benefits, harms, values, costs, equity, acceptability, and feasibility JAMA/WHO guideline PDF copy .
Conflict-of-interest management	WHO required declarations of interest. For the GLP-1 draft consultation, WHO reported 16 full responses accompanied by declarations of interest WHO guideline PDF via MAGICapp .
Public consultation	The GLP-1 draft guideline was opened for public consultation in September–October 2025 WHO consultation page . WHO’s adolescent obesity guideline was still in draft consultation in April 2026 and was not yet a final guideline as of the evidence cutoff WHO public comments call .
Equity and human rights	WHO emphasizes equity, feasibility, affordability, and person-centered care, especially for GLP-1 implementation and adolescent-obesity guideline development WHO news release , WHO public comments call .

II. Evidence Summary After WHO Latest Guideline Release (Evidence Cutoff Date: 29 April 2026)

Latest final WHO obesity-related guideline release used as baseline: WHO guideline on the use of GLP-1 therapies for the treatment of obesity in adults, published 1 December 2025.

2.1 New Evidence Overview

Search Strategy

Element	Description
Evidence window	2 December 2025 to 29 April 2026
Priority sources	WHO, PAHO/WHO, PubMed-indexed articles, JAMA, Nature Medicine, NEJM-trial summaries, Cochrane-cited reviews, FDA safety communications, clinical-practice guideline updates, systematic reviews, meta-analyses, and trial reports
Included evidence	Adult obesity pharmacotherapy evidence; GLP-1/GIP comparative data; safety alerts; real-world evidence; guideline-development updates for adolescent obesity

Element	Description
Excluded evidence	Non-obesity WHO guidelines except where relevant to youth engagement methodology; final adolescent obesity recommendations because WHO had not published a final guideline by the cutoff date
Main outcomes	Percent weight loss, categorical weight loss, glycaemic outcomes, diabetes remission/prevention, MASH/MASLD outcomes, obstructive sleep apnea outcomes, cardiovascular outcomes, adverse events, equity/implementation considerations

New evidence themes after 1 December 2025

1. **GLP-1/GIP pharmacotherapy evidence continued to strengthen**, especially for tirzepatide and semaglutide.
2. **Tirzepatide generally produced greater weight loss than semaglutide** in direct and indirect adult comparisons.
3. **Long-term treatment remains necessary**; discontinuation is associated with weight regain.
4. **Safety and quality concerns increased around compounded or unapproved GLP-1 products**, including dosing errors and serious gastrointestinal adverse events [FDA](#).
5. **Adolescent obesity guidance remained unfinished** as of the cutoff. WHO opened public consultation on draft adolescent obesity guidelines in April 2026, but final GRADE recommendations were not available [WHO public comments call](#).

2.2 Key New Studies

Study / Source	Design	Population	Key Findings	GRADE Quality	Clinical Interpretation
2026 systematic review and meta-analysis of pharmacological treatments for obesity in adults ; posted 24 April 2026 medRxiv full text	Systematic review and meta-analysis; preprint; 62 RCTs and 66 comparisons	Adults with overweight or obesity	All obesity medications reduced body weight more than placebo. Comparative hierarchy favored tirzepatide > semaglutide > liraglutide > orlistat for total body-weight loss PDF .	Moderate for weight-loss direction; downgraded for preprint status and heterogeneity	Supports WHO's conditional GLP-1 recommendation but does not justify making it strong because long-term safety, equity, cost, and implementation remain uncertain.
Active-comparator evidence	RCT active-comparator synthesis	Adults with obesity	Liraglutide outperformed orlistat;	Moderate	In adults eligible for pharmacotherapy,

Study / Source	Design	Population	Key Findings	GRADE Quality	Clinical Interpretation
summarized in 2026 meta-analysis PDF			semaglutide outperformed liraglutide; tirzepatide outperformed semaglutide. Tirzepatide vs semaglutide difference: 6.5% greater total body-weight loss , 95% CI 4.9–8.1, p<0.001.		choice of agent should consider efficacy, comorbidities, tolerability, contraindications, availability, cost, and patient preference.
SURMOUNT-5 evidence summarized in ICER and 2026 reviews ICER draft report PDF , Frontiers in Medicine	Head-to-head RCT evidence summarized in 2026 review context	Adults with obesity or overweight	At 72 weeks, tirzepatide produced greater adjusted weight loss than semaglutide: –20.2% vs –13.7% ; difference 6.5 percentage points . ≥20% weight loss: 48% vs 27% .	Moderate	Tirzepatide may be preferred when maximal weight loss is a priority and access/tolerability permit, but WHO's conditional framing remains appropriate.
Frontiers in Medicine narrative review , published 22 April 2026 Frontiers in Medicine	Narrative review of head-to-head GLP-1/GIP evidence	Adults with type 2 diabetes, obesity, or overweight	Identified SURPASS-2 and SURMOUNT-5 as the principal direct tirzepatide-vs-semaglutide randomized comparisons.	Low	Useful synthesis but not a replacement for systematic review; confirms direction of evidence favoring tirzepatide for weight loss.
Real-world comparative	Retrospective cohort	Adults without diabetes	Tirzepatide initiation was associated with	Low	Supports external validity of RCT findings but is

Study / Source	Design	Population	Key Findings	GRADE Quality	Clinical Interpretation
effectiveness study PMC		initiating obesity treatment	greater 6-month weight reduction and cardiometabolic improvements than semaglutide.		observational and susceptible to confounding.
FDA safety communication, current 4 February 2026 FDA	Regulatory safety communication	People using unapproved or compounded GLP-1-related products	FDA warned about illegally sold unapproved semaglutide, tirzepatide, or retatrutide products and adverse events from compounded products or non-label dosing; symptoms included nausea, vomiting, diarrhea, abdominal pain, constipation, and serious events.	Moderate for safety signal; regulatory evidence	Strongly supports avoiding compounded or unapproved GLP-1 products when approved regulated medicines are available.
WHO adolescent obesity draft guideline consultation, published 10 April 2026, comments due 24 April 2026 WHO public comments call	Guideline-development update, not final recommendations	Adolescents aged 10–19 years with obesity	WHO described draft guidance on integrated management using a primary health care approach, grounded in equity, gender, and human rights. Final recommendation	Not applicable; final GRADE unavailable	Do not extrapolate adult GLP-1 recommendations wholesale to adolescents; await final WHO adolescent guideline.

Study / Source	Design	Population	Key Findings	GRADE Quality	Clinical Interpretation
			wording and GRADE ratings were not public by 29 April 2026.		
Meaningful youth engagement evidence brief and 2026 case-study abstract WHO webinar , Journal of Pediatric and Adolescent Gynecology listing	Guideline-methodology case study; abortion-care guideline, not obesity	Adolescents and young people	WHO highlighted youth participation as a way to strengthen dialogue, policy relevance, inclusivity, quality, and legitimacy.	Low for transferability to obesity guideline development	Future adolescent-obesity guidelines should include meaningful youth engagement beyond public consultation.

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Priority research need	GRADE implication
Long-term GLP-1/GIP safety beyond trial duration	Obesity is chronic and relapsing; treatment may be lifelong	Longer RCT extensions and registries assessing pancreatobiliary, gastrointestinal, psychiatric, renal, pregnancy, sarcopenia, and nutritional outcomes	Current certainty remains Moderate rather than High
Weight regain after discontinuation	Evidence from STEP, SURMOUNT, and SCALE extensions indicates significant regain after stopping therapy Nature Medicine	Trials of maintenance dosing, tapering, behavioral maintenance, and combination care	Supports conditional long-term-treatment recommendations

Evidence gap	Why it matters	Priority research need	GRADE implication
Comparative effectiveness across diverse populations	Trials may underrepresent low-income settings, racial/ethnic groups, older adults, and people with multimorbidity	Pragmatic trials in primary care and low-resource settings	Equity and feasibility limitations reduce recommendation strength
Cost-effectiveness and equitable access	WHO emphasized need for transparent prioritization so highest-need groups are reached first WHO news release	Country-level cost-effectiveness and budget-impact studies	Resource uncertainty supports conditional implementation
Adolescents with obesity	WHO adolescent guideline was still draft as of April 2026 WHO public comments call	Age-specific trials of behavioral, pharmacologic, digital, sleep, physical activity, and surgical interventions	Adult evidence should not automatically be applied
Non-sugar sweeteners and long-term obesity outcomes	Short-term RCTs and long-term observational findings conflict WHO non-sugar sweeteners guideline	Long-term randomized and mechanistic studies	Current evidence remains Low
Lifestyle intervention intensity and sustainability	Behavioral therapy is recommended but low-certainty as a co-intervention with GLP-1 therapy	Trials comparing intensity, duration, digital support, dietitian-led models, and culturally tailored approaches	Current co-intervention recommendation remains Conditional
Meaningful youth engagement in adolescent obesity guidelines	Public consultation does not necessarily equal youth co-development	Include adolescents as equal partners across scoping, evidence interpretation, recommendation formulation, implementation, and dissemination	Methodological certainty currently Low but ethically important

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact Through 29 April 2026	Updated Recommendation
Adults with obesity may use GLP-1 therapies as long-term treatment; conditional ,	2026 evidence supports clinically meaningful weight loss, especially with tirzepatide and semaglutide,	Maintain conditional recommendation. Offer approved GLP-1/GIP therapy to eligible adults after

Original WHO Recommendation	New Evidence Impact Through 29 April 2026	Updated Recommendation
moderate-certainty evidence JAMA/WHO guideline PDF copy	but includes preprint evidence and unresolved equity/cost issues medRxiv full text .	shared decision-making, comorbidity assessment, contraindication review, and access/cost discussion.
Intensive behavioral therapy may be provided with GLP-1 therapy; conditional, low-certainty evidence JAMA/WHO guideline PDF copy	No strong post-release evidence overturns the low-certainty rating. Behavioral care remains important for diet quality, physical activity, adherence, and weight-loss maintenance.	Maintain conditional recommendation. Pair pharmacotherapy with nutrition counseling, physical activity planning, behavioral support, sleep assessment, and long-term follow-up whenever feasible.
Obesity should be treated as chronic disease requiring comprehensive care PAHO/WHO	Evidence of weight regain after stopping medication reinforces need for chronic-care models Nature Medicine .	Strengthen implementation emphasis. Avoid short-term “weight-loss course” framing; plan long-term monitoring and maintenance.
Use approved pharmacotherapy within safe systems	FDA warnings highlight risks from unapproved and compounded GLP-1 products FDA .	Do not prescribe or recommend unapproved, illegally marketed, or non-standard compounded GLP-1 products when regulated approved products are available. Monitor for gastrointestinal and other adverse effects.
Dietary prevention: total fat ≤30%, saturated fat ≤10%, trans fat ≤1%, free sugars <10%, avoid reliance on NSS	No post-release evidence changes WHO dietary guidance. Pharmacotherapy does not replace healthy-diet policy.	Maintain WHO dietary recommendations. Medication should not substitute for dietary-quality improvement, food-system measures, and physical activity promotion.
Adolescent obesity management	WHO adolescent obesity guideline remained draft as of 29 April 2026; final GRADE recommendations unavailable WHO public comments call .	Do not treat adult GLP-1 guidance as adolescent guidance. Use existing pediatric standards and await final WHO adolescent guideline; include adolescents and families in shared decision-making.
Youth engagement in guideline development	WHO youth-engagement case study was for abortion care, not obesity, but supports inclusion of youth voices WHO webinar .	Future adolescent obesity guidance should include meaningful youth engagement , not only public comment, especially for acceptability, stigma, feasibility, and implementation.

Practical WHO-Based Care Pathway for Obesity

Step	Action	WHO-based rationale
1	Screen and diagnose using BMI and age-specific WHO growth standards; assess waist circumference or central adiposity where relevant.	WHO BMI thresholds define adult and pediatric overweight/obesity WHO BMI-for-age reference .
2	Assess complications and context: type 2 diabetes, cardiovascular disease, sleep apnea, MASLD/MASH, osteoarthritis, mental health, medications, pregnancy potential, stigma, food insecurity, and social determinants.	WHO frames obesity as chronic, progressive, and relapsing PAHO/WHO .
3	Start comprehensive lifestyle and behavioral care: healthy diet, physical activity, sleep, reduction of sedentary behavior, behavioral support, and family/community supports where appropriate.	WHO emphasizes GLP-1 therapy as part of comprehensive care, not standalone treatment WHO news release .
4	Apply WHO diet targets: total fat $\leq 30\%$ energy; saturated fat $\leq 10\%$; trans fat $\leq 1\%$; free sugars $< 10\%$, preferably $< 5\%$ where feasible; avoid using NSS for long-term weight control.	WHO dietary guidelines address unhealthy weight gain through overall diet quality and energy balance WHO news , WHO non-sugar sweeteners guideline .
5	Consider pharmacotherapy in eligible adults after shared decision-making.	WHO conditionally recommends long-term GLP-1 therapy for adults with obesity JAMA/WHO guideline PDF copy .
6	Choose medication based on patient factors: expected efficacy, comorbidities, adverse effects, contraindications, cost, availability, route, preference, and equity.	New evidence suggests tirzepatide and semaglutide produce the largest weight reductions, but implementation constraints remain medRxiv full text .
7	Avoid unapproved or unsafe products.	FDA warned about unapproved GLP-1-related products and adverse events from compounded or non-label dosing FDA .
8	Monitor long term: weight trajectory, waist/central adiposity, cardiometabolic markers, adverse effects, nutrition status, mental health, adherence, access, and patient goals.	Long-term therapy and monitoring are needed because obesity is chronic and weight regain after discontinuation is common Nature Medicine .

Overall conclusion: As of **29 April 2026**, WHO-based obesity management should combine accurate diagnosis, chronic disease framing, healthy-diet policy, physical activity and behavioral support, and—where appropriate for adults—approved long-term GLP-1/GIP pharmacotherapy. The newest post-WHO evidence strengthens confidence that tirzepatide and semaglutide are highly effective for adult weight loss, but it does **not** remove the need for conditional, equity-aware, long-term, multimodal care.